

WASP-104b

R-2586

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_160324 · created 2026-08-27T22:49:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457471.6428 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.041
Transit depth (Rp/Rs)^2	1.83 +/- 1.11 [%]
Orbital Inclination (inc)	82.81 +/- 1.5
Airmass coefficient 1 (a1)	1.115 +/- 0.018
Airmass coefficient 2 (a2)	-0.096 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	None
Optimal Aperture	3.38
Optimal Annulus	13.03
Transit Duration (day)	0.052 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.041 vs 0.1214 (deviation 0.18σ)
Duration fit vs published	0.052 d vs 0.0742 d (deviation 0.84σ)
Mid-time O–C	-0.5 min
Depth SNR	1.7
Residual scatter	1.6 %

Light curve

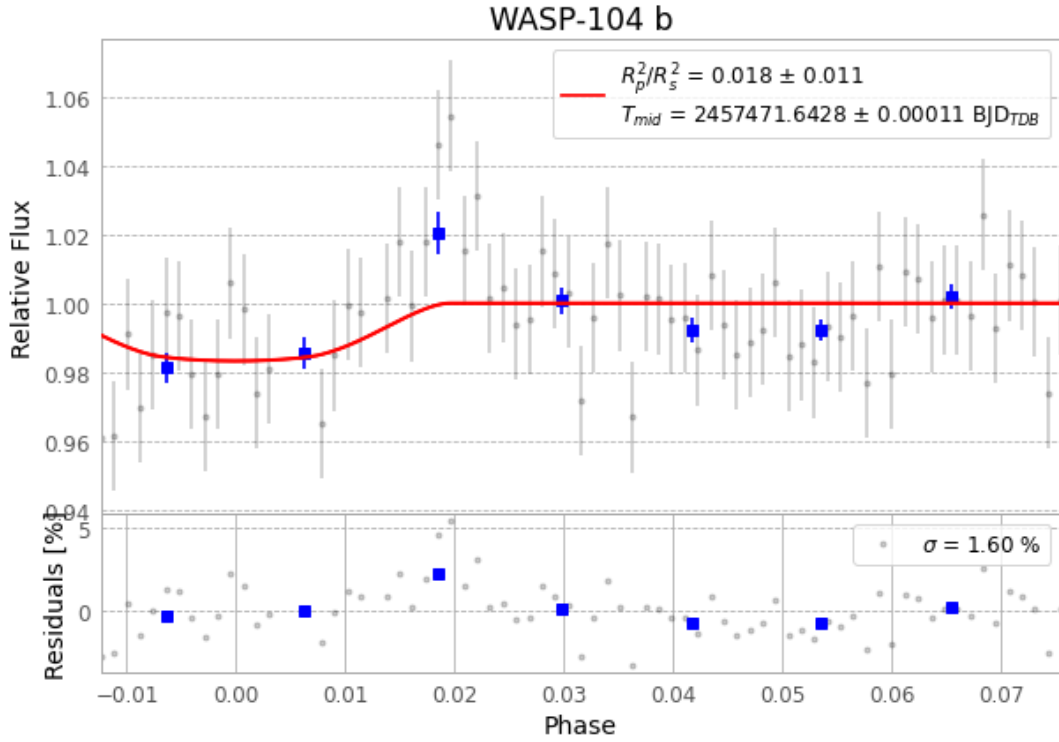


Figure 1 — FinalLightCurve_WASP-104 b_2016-03-23.png (SHA-256 991370d731aee08f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0591c1774e57c3e9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), WASP-104160324024512.FITS → WASP-104160324062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160324.log (8f29d76506e9...), FinalLightCurve_WASP-104 b_2016-03-23.png (991370d731ae...), FinalLightCurve_WASP-104 b_2016-03-23.csv (2159e5339593...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 22:49Z.